Investments

1 Dave's investment of $\$6000$ earns interest at 2% p.a, compounded annually over 3 years.

Using repeated multiplication, find:

a The value of the investment after 3 years.

b The amount of interest earned.

2 A $\$2090$ investment earns interest at 4.2% p.a. compounded annually over 17 years. Use the compound interest formula to calculate the value of this investment.

3 Sally's investment of $\$8950$ earns interest at 4% p.a. compounded annually over 4 years. Find the amount of interest earned.

4 $\$3900$ is invested for three years at a rate of 10% p.a., compounding annually.

a Complete the table below to determine the final value of the investment:

	Balance + interest	Total balance	Interest earned
First year	–	$\$3900$	$\$390$
Second year	$\$3900 + \$390$	$\$4290$	$\$429$
Third year			
Fourth year			–

b Calculate the total interest earned over the three years.

5 State whether the following is true about compound interest:

a Interest is earned on the principal.

b The interest in any time period is calculated using only the original principal.

c Interest is earned on any accumulated interest.

d The amount of interest earned in any time period changes from one period to the next.

Inflation

6 What is the value of a $\$1330$ brand new desk after 7 years when the annual inflation rate is 11%?

7 In 2001, the price of a bottle of orange juice was $\$3.20$. What was the price of a bottle of

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8 For an inflation rate of 7.4% per year, what is the expected value of a $\$210$ basket of groceries 7 years from now?

9 A one year sports club membership currently costs $\$332$. Calculate the cost in 6 years time if the inflation rate is 2.6% per annum. Round your answer to the nearest dollar.

10 In terms of today's dollar, what will be the value of $\$5$ in 15 years time if the inflation rate is:

a 2.1%

b 4.1%

11 Find the original price of the following items:

a A hat today costs $\$35.50$. Find the price of the hat 9 years ago at an inflation rate of 3.7% per annum.

b An item costs $\$15\,000$ today. Find the price of the item 4 years ago at an inflation rate of 3.5% per annum.

12 The price of pens and pencils in 1996 was $\$5$. If the value inflated at an average rate of 3.2% per annum, what would the price have been in 2005?

Appreciation

13 What is the value of a wine collection, currently valued at $\$4790$, after 10 years if its rate of appreciation is 9% p.a.?

14 The value of land is expected to grow by 6.8% each year for the next 5 years. If it currently costs $\$460$ per square metre, how much will it cost at the end of 5 years?

15 Due to dwindling supplies of oil, oil prices are expected to appreciate by 7.2% p.a. If the current price of oil is $\$79$ a barrel, what is the expected price of a barrel in 23 years time?

16 A house was valued 6 years ago to be worth $\$548\,000$. Its value appreciated at 5.2% p.a. What is its appreciated value? Give your answer to the nearest dollar.

17 If a piece of land appreciates at an average rate of 3.7% per annum and its current value is $\$430\,000$, calculate its value in 3 years. Give your answer to the nearest dollar.

18 A high interest bank account deposit appreciates at 6.6% per annum and is currently valued at $\$46\,000$. What will its value be in 8 years? Give your answer to the nearest dollar.

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19 Sean deposits $\$4000$ into a new savings account. The amount increased by 3.3% each year for 4 years. Find the new value of the deposit after appreciation, giving your answer to the nearest dollar.

20 The federal government wants to increase funding to a certain program over 8 years at a constant rate of 4.75% per annum over that time period. If the initial funding was $\$14\,000\,000$, find the value of the funding (to the nearest dollar) after:

a 1 year **b** 4 years **c** 8 years

21 A vintage collectors item that costs $\$6000$, appreciates at approximately 6.6% p.a. After how many full years, n, will the value of the vintage collectors item be over $\$15\,000$?

Depreciation

22 There were $10\,100$ digital sales of a particular song in its first month of release. The number of monthly digital sales decreased by 360 each month after.

a The song registered 7580 digital sales in a particular month. What was the decrease in the monthly sales from the first month?

b In how many months after the initial month of release did the song register 7580 in monthly digital sales?

23 Production robots to be used in a car manufacturing plant were purchased for $\$4\,455\,000$. After 5 years, they depreciated to a value of $\$4\,385\,000$.

a What was the annual depreciation using the straight-line method?

b After 7 years, the robots are sold off. If they continue to depreciate at an annual rate of $\$14\,000$, how much can they be sold for?

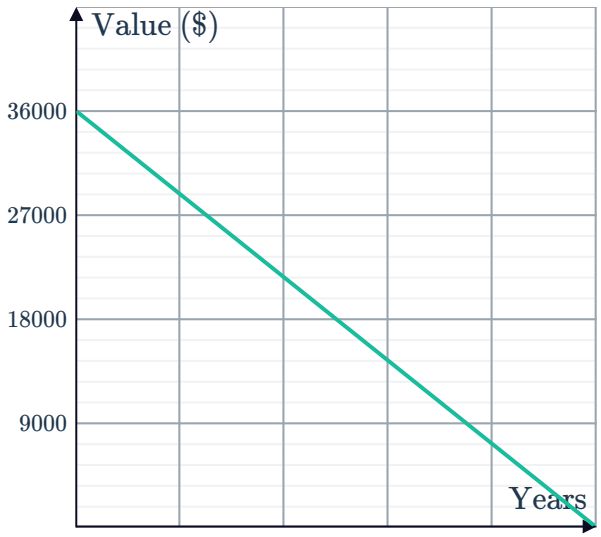
24 The following graph shows the depreciation of a car's value over 4 years:

a What is the initial value of the car?

b By how much did the car depreciate each year?

c After how many years will the car be worth $\$14\,400$?

d What is the value of the car after 4 years?



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25 Marge purchased shares at a total value of $\$99\,800$. After 6 months, the shares had dropped in value to $\$99\,200$.

a What is the monthly depreciation using the straight-line method?

b Sketch the graph showing the depreciation of the value of the shares over 6 years.

c What is the gradient of the line?

d What does the gradient represent in context?

e What is the value of the y-intercept of the line?

f What does the y-intercept represent in context?

g Let V represent the value of the shares after n months. Write the equation of the line.

h Find the value of the shares after $5\frac{1}{2}$ years.

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